Lodos's mj mode against MuJoCo Warp: batched throughput

2026-11-02

scene	s	Δt (ms)	steps	solref (ms)	Lodos's solver
cube_walled_5_polytope	3	2	1500	5	100 it, tol 1e-06, ls 50 it
cube_walled_5_sdf	3	2	1500	5	100 it, tol 1e-06, ls 50 it
arm_walled_5_polytope	3	2	1500	5	100 it, tol 1e-06, ls 50 it
arm_walled_5_sdf	3	2	1500	5	100 it, tol 1e-06, ls 50 it
shadow_grasp_polytope	5	2	2500	5	100 it, tol 1e-06, ls 50 it
shadow_grasp_sdf	5	2	2500	5	100 it, tol 1e-06, ls 50 it
shadow_walled_5_polytope	3	2	1500	5	100 it, tol 1e-06, ls 50 it
shadow_walled_5_sdf	3	2	1500	5	100 it, tol 1e-06, ls 50 it
nist_threading_m8	5	1.25	4000	2.5	100 it, tol 1e-05, ls 50 it
cube_walled_20_polytope	3	2	1500	5	100 it, tol 1e-06, ls 50 it
cube_walled_20_sdf	3	2	1500	5	100 it, tol 1e-06, ls 50 it
arm_walled_20_polytope	3	2	1500	5	100 it, tol 1e-06, ls 50 it
arm_walled_20_sdf	3	2	1500	5	100 it, tol 1e-06, ls 50 it
shadow_walled_20_polytope	3	2	1500	5	100 it, tol 1e-06, ls 50 it
shadow_walled_20_sdf	3	2	1500	5	100 it, tol 1e-06, ls 50 it

Table 1: The scenes: the simulated seconds a run times, the step, the steps, every contact's solref time constant (damping ratio 1), and the solver settings both sims run (the viewer's defaults for Lodos's mj mode).

scene	configuration	a world's capacities (1.5 times its peak over 16 starts)
cube_walled_5_polytope	Lodos, contact	nconmax 30, njmax 90, nccdmax 12
cube_walled_5_polytope	MuJoCo Warp	contacts 53, triangle pairs 1, rows 104
cube_walled_5_sdf	Lodos, contact	nconmax 141, njmax 423; capacity_tile 0.103, capacity2 0.395
cube_walled_5_sdf	Lodos, cone	nconmax 18, njmax 54; capacity_tile 0.106, capacity2 0.373
cube_walled_5_sdf	Lodos, ray	nconmax 18, njmax 18; capacity_tile 0.103, capacity2 0.387
cube_walled_5_sdf	MuJoCo Warp	contacts 257, triangle pairs 1, rows 581
arm_walled_5_polytope	Lodos, contact	nconmax 228, njmax 104, nccdmax 21
arm_walled_5_polytope	MuJoCo Warp	contacts 255, triangle pairs 1, rows 171
arm_walled_5_sdf	Lodos, contact	nconmax 24, njmax 72; capacity_tile 0.163, capacity2 0.0351
arm_walled_5_sdf	Lodos, cone	nconmax 21, njmax 63; capacity_tile 0.161, capacity2 0.0372
arm_walled_5_sdf	Lodos, ray	nconmax 23, njmax 23; capacity_tile 0.178, capacity2 0.0491
arm_walled_5_sdf	MuJoCo Warp	contacts 273, triangle pairs 1, rows 95
shadow_grasp_polytope	Lodos, contact	nconmax 56, njmax 123, nccdmax 14
shadow_grasp_polytope	MuJoCo Warp	contacts 21, triangle pairs 1, rows 74
shadow_grasp_sdf	Lodos, contact	nconmax 14, njmax 78; capacity_tile 0.0347, capacity2 0.0675
shadow_grasp_sdf	Lodos, cone	nconmax 8, njmax 60; capacity_tile 0.0347, capacity2 0.0675
shadow_grasp_sdf	Lodos, ray	nconmax 8, njmax 45; capacity_tile 0.0562, capacity2 0.0523
shadow_grasp_sdf	MuJoCo Warp	contacts 233, triangle pairs 1, rows 699
shadow_walled_5_polytope	Lodos, contact	nconmax 698, njmax 842, nccdmax 63

scene	configuration	a world's capacities (1.5 times its peak over 16 starts)
shadow_walled_5_polytope	MuJoCo Warp	contacts 252, triangle pairs 1, rows 566
shadow_walled_5_sdf	Lodos, contact	nconmax 330, njmax 1211; capacity_tile 0.0218, capacity2 0.0423
shadow_walled_5_sdf	Lodos, cone	nconmax 56, njmax 408; capacity_tile 0.0199, capacity2 0.0427
shadow_walled_5_sdf	Lodos, ray	nconmax 65, njmax 303; capacity_tile 0.0219, capacity2 0.0767
shadow_walled_5_sdf	MuJoCo Warp	contacts 1119, triangle pairs 1, rows 3515
nist_threading_m8	Lodos, contact	nconmax 110, njmax 329, nccdmax 2; capacity_tile 0.251, capacity2 0.249
nist_threading_m8	Lodos, cone	nconmax 26, njmax 77, nccdmax 2; capacity_tile 0.253, capacity2 0.243
nist_threading_m8	Lodos, ray	nconmax 42, njmax 45, nccdmax 2; capacity_tile 0.454, capacity2 0.181
nist_threading_m8	MuJoCo Warp	contacts 423, triangle pairs 5294, rows 366
cube_walled_20_polytope	Lodos, contact	nconmax 111, njmax 333, nccdmax 24
cube_walled_20_polytope	MuJoCo Warp	contacts 168, triangle pairs 1, rows 360
cube_walled_20_sdf	Lodos, contact	nconmax 1136, njmax 3407; capacity_tile 0.0311, capacity2 0.362
cube_walled_20_sdf	Lodos, cone	nconmax 45, njmax 135; capacity_tile 0.0307, capacity2 0.324
cube_walled_20_sdf	Lodos, ray	nconmax 51, njmax 51; capacity_tile 0.0332, capacity2 0.378
cube_walled_20_sdf	MuJoCo Warp	contacts 810, triangle pairs 1, rows 1944
arm_walled_20_polytope	Lodos, contact	nconmax 839, njmax 293, nccdmax 54
arm_walled_20_polytope	MuJoCo Warp	contacts 684, triangle pairs 1, rows 414
arm_walled_20_polytope	MuJoCo Warp, SAP	contacts 755, triangle pairs 1, rows 437, shape pairs 2769
arm_walled_20_sdf	Lodos, contact	nconmax 87, njmax 261; capacity_tile 0.0553, capacity2 0.0349
arm_walled_20_sdf	Lodos, cone	nconmax 69, njmax 207; capacity_tile 0.0583, capacity2 0.0337
arm_walled_20_sdf	Lodos, ray	nconmax 65, njmax 65; capacity_tile 0.0576, capacity2 0.035
arm_walled_20_sdf	MuJoCo Warp	contacts 939, triangle pairs 1, rows 270
shadow_walled_20_polytope	Lodos, contact	nconmax 2537, njmax 2208, nccdmax 134
shadow_walled_20_polytope	MuJoCo Warp	contacts 893, triangle pairs 1, rows 1938
shadow_walled_20_polytope	MuJoCo Warp, SAP	contacts 866, triangle pairs 1, rows 1926, shape pairs 10058
shadow_walled_20_sdf	Lodos, contact	nconmax 374, njmax 1991; capacity_tile 0.00457, capacity2 0.0284
shadow_walled_20_sdf	Lodos, cone	nconmax 186, njmax 1526; capacity_tile 0.00466, capacity2 0.0304
shadow_walled_20_sdf	Lodos, ray	nconmax 218, njmax 1185; capacity_tile 0.00491, capacity2 0.0437
shadow_walled_20_sdf	MuJoCo Warp	contacts 3992, triangle pairs 81326, rows 12653

Table 2: Each configuration's capacities per world, times the worlds at a batch size: MuJoCo Warp's contacts, rows and CCD slots, Lodos's arenas as fractions of a world's worst case; Newton's contacts and mesh triangle pairs (at least a million in all) and rows, and under its sweep-and-prune broad phase (SAP) its candidate shape pairs.

scene	Lodos, contact	Lodos, cone	Lodos, ray	MuJoCo Warp
cube_walled_5_polytope	20 / 60	n/a	n/a	35 / 69
cube_walled_5_sdf	94 / 282	12 / 36	12 / 12	171 / 387
arm_walled_5_polytope	152 / 69	n/a	n/a	170 / 114
arm_walled_5_sdf	16 / 48	14 / 42	15 / 15	182 / 63

scene	Lodos, contact	Lodos, cone	Lodos, ray	MuJoCo Warp
shadow_grasp_polytope	37 / 82	n/a	n/a	14 / 49
shadow_grasp_sdf	9 / 52	5 / 40	5 / 30	155 / 466
shadow_walled_5_polytope	465 / 561	n/a	n/a	168 / 377
shadow_walled_5_sdf	220 / 807	37 / 272	43 / 202	746 / 2343
nist_threading_m8	73 / 219	17 / 51	28 / 30	282 / 244
cube_walled_20_polytope	74 / 222	n/a	n/a	112 / 240
cube_walled_20_sdf	757 / 2271	30 / 90	34 / 34	540 / 1296
arm_walled_20_polytope	559 / 195	n/a	n/a	456 / 276
arm_walled_20_sdf	58 / 174	46 / 138	43 / 43	626 / 180
shadow_walled_20_polytope	1691 / 1472	n/a	n/a	595 / 1292
shadow_walled_20_sdf	249 / 1327	124 / 1017	145 / 790	2661 / 8435

Table 3: A world's largest demand over the 16 starts: contacts / constraint rows of the contact solve (Lodos's: MJWarp's contacts and rows, its surfaces' contacts among them, a group of them a contact). n/a: a polytope scene has no Lodos surfaces, so no groups.

scene	groups	B	default	no shared H.	fused Newton	kept
cube_walled_5_polytope	contact	1024	965k	1.07M	–	no shared H.
cube_walled_5_sdf	contact	1024	583k	601k	–	no shared H.
cube_walled_5_sdf	cone	1024	868k	871k	–	default
cube_walled_5_sdf	ray	1024	1.27M	1.29M	–	default
arm_walled_5_polytope	contact	1024	683k	685k	–	default
arm_walled_5_sdf	contact	1024	923k	932k	–	default
arm_walled_5_sdf	cone	1024	888k	892k	–	default
arm_walled_5_sdf	ray	1024	1.33M	1.41M	–	no shared H.
shadow_grasp_polytope	contact	1024	688k	706k	–	default
shadow_grasp_sdf	contact	1024	825k	914k	–	no shared H.
shadow_grasp_sdf	cone	1024	759k	864k	–	no shared H.
shadow_grasp_sdf	ray	1024	896k	944k	–	no shared H.
shadow_walled_5_polytope	contact	1024	148k	138k	–	default
shadow_walled_5_sdf	contact	1024	107k	101k	–	default
shadow_walled_5_sdf	cone	1024	94.1k	91.3k	–	default
shadow_walled_5_sdf	ray	1024	78.7k	74.3k	–	default
nist_threading_m8	contact	1024	667k	678k	888k	fused Newton
nist_threading_m8	cone	1024	781k	788k	951k	fused Newton
nist_threading_m8	ray	1024	790k	789k	941k	fused Newton
cube_walled_20_polytope	contact	256	146k	131k	–	default
cube_walled_20_sdf	contact	256	87.6k	83k	–	default
cube_walled_20_sdf	cone	256	101k	90.2k	–	default
cube_walled_20_sdf	ray	256	138k	128k	–	default
arm_walled_20_polytope	contact	256	94.8k	88.4k	–	default
arm_walled_20_sdf	contact	256	120k	110k	–	default
arm_walled_20_sdf	cone	256	105k	95.5k	–	default
arm_walled_20_sdf	ray	256	150k	140k	–	default
shadow_walled_20_polytope	contact	256	4.11k	4.1k	–	default
shadow_walled_20_sdf	contact	256	3.22k	3.21k	–	default
shadow_walled_20_sdf	cone	256	2.04k	2.04k	–	default

scene	groups	B	default	no shared H.	fused Newton	kept
shadow_walled_20_sdf	ray	256	1.78k	1.78k	–	default

Table 4: Lodos's execution options, world-steps/s at B worlds, from the capacity runs' half-way states: the default with MJWarp's Hessian summed in shared memory (on by default where it builds), without it (no shared H.), and the fused Newton solve; an option is kept where it is 3 % faster ("–": refused, the fused solve takes models of at most 16 degrees of freedom).

scene	Lodos, contact	Lodos, cone	Lodos, ray	MuJoCo Warp	ratio
cube_walled_5_polytope	1.61M @8192	n/a	n/a	1.11M @8192	1.45
cube_walled_5_sdf	551k @4096	1.27M @8192	1.86M @8192	276k @2048	6.76
arm_walled_5_polytope	985k @8192	n/a	n/a	163k @2048	6.04
arm_walled_5_sdf	1.24M @4096	1.25M @8192	1.97M @8192	238k @2048	8.29
shadow_grasp_polytope	1.18M @8192	n/a	n/a	432k @32768	2.73
shadow_grasp_sdf	1.38M @8192	1.25M @8192	1.42M @8192†	126k @4096†	10.94
shadow_walled_5_polytope	130k @8192	n/a	n/a	36.2k @2048	3.60
shadow_walled_5_sdf	112k @4096	113k @8192	110k @8192	17.4k @4096	6.46
nist_threading_m8	1.74M @8192	1.94M @8192	1.97M @8192†	52k @1024	37.35
cube_walled_20_polytope	394k @16384	n/a	n/a	256k @16384	1.54
cube_walled_20_sdf	119k @4096	263k @16384	349k @8192	107k @4096	3.25
arm_walled_20_polytope	178k @16384	n/a	n/a	46.3k @2048§	3.85
arm_walled_20_sdf	261k @16384	259k @16384	366k @16384	84k @4096	4.36
shadow_walled_20_polytope	3.31k @512	n/a	n/a	2.59k @512§	1.28
shadow_walled_20_sdf	2.88k @512	2.3k @512	2.36k @512	626 @64	4.60

Table 5: The best throughput of each configuration, world-steps/s at the batch size after @, and the ratio of Lodos's to MuJoCo Warp's, Lodos's the fastest of its configurations that lost no world, MuJoCo Warp's the faster of its broad phases where it ran both. †: the run lost worlds, non-finite or escaped (Table 7). §: under Newton's sweep-and-prune broad phase (Table 6). n/a: a polytope scene has no Lodos surfaces, so no groups.

scene	configuration	world-steps/s	B	× real time	× MuJoCo Warp
cube_walled_5_polytope	Lodos, contact	1.61M	8192	3 230	1.45
	MuJoCo Warp	1.11M	8192	2 230	1.00
cube_walled_5_sdf	Lodos, contact	551k	4096	1 100	2.00
	Lodos, cone	1.27M	8192	2 550	4.62
	Lodos, ray	1.86M	8192	3 720	6.76
	MuJoCo Warp	276k	2048	551	1.00
arm_walled_5_polytope	Lodos, contact	985k	8192	1 970	6.04
	MuJoCo Warp	163k	2048	326	1.00
arm_walled_5_sdf	Lodos, contact	1.24M	4096	2 490	5.22
	Lodos, cone	1.25M	8192	2 500	5.25
	Lodos, ray	1.97M	8192	3 950	8.29
	MuJoCo Warp	238k	2048	476	1.00
shadow_grasp_polytope	Lodos, contact	1.18M	8192	2 350	2.73
	MuJoCo Warp	432k	32768	864	1.00
shadow_grasp_sdf	Lodos, contact	1.38M	8192	2 770	10.94
	Lodos, cone	1.25M	8192	2 510	9.92
	Lodos, ray	1.42M†	8192	2 830	11.19
	MuJoCo Warp	126k†	4096	253	1.00

scene	configuration	world-steps/s	B	× real time	× MuJoCo Warp
`shadow_walled_5_polytope`	Lodos, contact	130k	8192	260	3.60
	MuJoCo Warp	36.2k	2048	72.4	1.00
`shadow_walled_5_sdf`	Lodos, contact	112k	4096	223	6.40
	Lodos, cone	113k	8192	226	6.46
	Lodos, ray	110k	8192	219	6.29
	MuJoCo Warp	17.4k	4096	34.9	1.00
`nist_threading_m8`	Lodos, contact	1.74M	8192	2 180	33.50
	Lodos, cone	1.94M	8192	2 430	37.35
	Lodos, ray	1.97M†	8192	2 460	37.89
	MuJoCo Warp	52k	1024	65	1.00
`cube_walled_20_polytope`	Lodos, contact	394k	16384	788	1.54
	MuJoCo Warp	256k	16384	512	1.00
`cube_walled_20_sdf`	Lodos, contact	119k	4096	239	1.11
	Lodos, cone	263k	16384	525	2.45
	Lodos, ray	349k	8192	698	3.25
	MuJoCo Warp	107k	4096	215	1.00
`arm_walled_20_polytope`	Lodos, contact	178k	16384	356	3.85
	MuJoCo Warp	44.2k	512	88.4	0.95
	MuJoCo Warp, SAP	46.3k	2048	92.6	1.00
`arm_walled_20_sdf`	Lodos, contact	261k	16384	522	3.11
	Lodos, cone	259k	16384	517	3.08
	Lodos, ray	366k	16384	732	4.36
	MuJoCo Warp	84k	4096	168	1.00
`shadow_walled_20_polytope`	Lodos, contact	3.31k	512	6.61	1.28
	MuJoCo Warp	1.56k	128	3.12	0.60
	MuJoCo Warp, SAP	2.59k	512	5.18	1.00
`shadow_walled_20_sdf`	Lodos, contact	2.88k	512	5.76	4.60
	Lodos, cone	2.3k	512	4.6	3.67
	Lodos, ray	2.36k	512	4.72	3.77
	MuJoCo Warp	626	64	1.25	1.00

Table 6: Each configuration's best throughput: world-steps/s at a batch of B worlds; how many times real time it ran, the seconds simulated per second summed over the worlds (world-steps/s times the step, 2 ms, the threading's 1.25 ms; each world's own is that over B); and its ratio to MuJoCo Warp's best on the scene, over its broad phases. MuJoCo Warp, SAP: under Newton's sweep-and-prune broad phase, its candidate shape pairs a capacity measured as the others, in place of the explicit pairs, whose narrow-phase buffers Newton sizes for every pair of shapes in every world. †: the run lost worlds, non-finite or escaped (Table 7).

scene	configuration	world-steps/s by batch size; how the ladder ended; worlds lost at the best
`cube_walled_5_polytope`	Lodos, contact	64: 111k, 128: 201k, 256: 358k, 512: 598k, 1024: 917k, 2048: 1.27M, 4096: 1.55M, 8192: 1.61M, 16384: 1.59M; dropped at 16384; 0 non-finite, 0 escaped
`cube_walled_5_polytope`	MuJoCo Warp	64: 113k, 128: 201k, 256: 344k, 512: 543k, 1024: 782k, 2048: 987k, 4096: 1.1M, 8192: 1.11M, 16384: 1.07M; dropped at 16384; 0 non-finite, 0 escaped

scene	configuration	world-steps/s by batch size; how the ladder ended; worlds lost at the best
cube_walled_5_sdf	Lodos, contact	64: 75.8k, 128: 133k, 256: 217k, 512: 328k, 1024: 444k, 2048: 532k, 4096: 551k, 8192: 536k; dropped at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow
cube_walled_5_sdf	Lodos, cone	64: 102k, 128: 186k, 256: 332k, 512: 560k, 1024: 805k, 2048: 1.05M, 4096: 1.21M, 8192: 1.27M, 16384: 1.25M; dropped at 16384; 0 non-finite, 0 escaped
cube_walled_5_sdf	Lodos, ray	64: 135k, 128: 258k, 256: 467k, 512: 794k, 1024: 1.17M, 2048: 1.54M, 4096: 1.78M, 8192: 1.86M, 16384: 1.83M; dropped at 16384; 0 non-finite, 0 escaped
cube_walled_5_sdf	MuJoCo Warp	64: 65.7k, 128: 107k, 256: 158k, 512: 211k, 1024: 257k, 2048: 276k, 4096: 266k; dropped at 4096; 0 non-finite, 0 escaped; 5 redone after an overflow
arm_walled_5_polytope	Lodos, contact	64: 93.9k, 128: 169k, 256: 290k, 512: 462k, 1024: 625k, 2048: 807k, 4096: 939k, 8192: 985k, 16384: 975k; dropped at 16384; 0 non-finite, 0 escaped; 1 redone after an overflow
arm_walled_5_polytope	MuJoCo Warp	64: 55.3k, 128: 85.5k, 256: 118k, 512: 146k, 1024: 160k, 2048: 163k, 4096: 162k; dropped at 4096; 0 non-finite, 0 escaped; 2 redone after an overflow
arm_walled_5_sdf	Lodos, contact	64: 118k, 128: 216k, 256: 383k, 512: 620k, 1024: 866k, 2048: 1.11M, 4096: 1.24M, 8192: 1.24M; dropped at 8192; 0 non-finite, 0 escaped
arm_walled_5_sdf	Lodos, cone	64: 113k, 128: 213k, 256: 365k, 512: 596k, 1024: 849k, 2048: 1.09M, 4096: 1.23M, 8192: 1.25M, 16384: 1.23M; dropped at 16384; 0 non-finite, 0 escaped
arm_walled_5_sdf	Lodos, ray	64: 164k, 128: 305k, 256: 528k, 512: 879k, 1024: 1.29M, 2048: 1.7M, 4096: 1.92M, 8192: 1.97M, 16384: 1.97M; dropped at 16384; 0 non-finite, 0 escaped
arm_walled_5_sdf	MuJoCo Warp	64: 81.7k, 128: 128k, 256: 171k, 512: 208k, 1024: 229k, 2048: 238k, 4096: 229k; dropped at 4096; 0 non-finite, 0 escaped; 1 redone after an overflow
shadow_grasp_polytope	Lodos, contact	64: 97.5k, 128: 179k, 256: 319k, 512: 527k, 1024: 745k, 2048: 1.01M, 4096: 1.14M, 8192: 1.18M, 16384: 1.15M; dropped at 16384; 0 non-finite, 0 escaped
shadow_grasp_polytope	MuJoCo Warp	64: 72.6k, 128: 129k, 256: 199k, 512: 287k, 1024: 349k, 2048: 393k, 4096: 414k, 8192: 422k, 16384: 428k, 32768: 432k, 65536: –; out of memory at 65536; 0 non-finite, 0 escaped
shadow_grasp_sdf	Lodos, contact	64: 116k, 128: 212k, 256: 369k, 512: 632k, 1024: 889k, 2048: 1.2M, 4096: 1.34M, 8192: 1.38M, 16384: 1.32M; dropped at 16384; 0 non-finite, 0 escaped
shadow_grasp_sdf	Lodos, cone	64: 93.9k, 128: 172k, 256: 303k, 512: 513k, 1024: 767k, 2048: 1.05M, 4096: 1.2M, 8192: 1.25M, 16384: 1.21M; dropped at 16384; 0 non-finite, 0 escaped
shadow_grasp_sdf	Lodos, ray	64: 101k, 128: 186k, 256: 330k, 512: 559k, 1024: 828k, 2048: 1.16M, 4096: 1.33M, 8192: 1.42M, 16384: 1.37M; dropped at 16384; 0 non-finite, 8192 escaped
shadow_grasp_sdf	MuJoCo Warp	64: 35.3k, 128: 55.8k, 256: 78.7k, 512: 102k, 1024: 119k, 2048: 124k, 4096: 126k, 8192: 125k; dropped at 8192; 0 non-finite, 1 escaped; 1 redone after an overflow
shadow_walled_5_polytope	Lodos, contact	64: 23.2k, 128: 38.7k, 256: 62.4k, 512: 87.5k, 1024: 109k, 2048: 121k, 4096: 128k, 8192: 130k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped; 1 redone after an overflow
shadow_walled_5_polytope	MuJoCo Warp	64: 16.8k, 128: 22.5k, 256: 29k, 512: 33.2k, 1024: 34.8k, 2048: 36.2k, 4096: –; out of memory at 4096; 0 non-finite, 0 escaped

scene	configuration	world-steps/s by batch size; how the ladder ended; worlds lost at the best
shadow_walled_5_sdf	Lodos, contact	64: 19.5k, 128: 29.4k, 256: 41.6k, 512: 65.2k, 1024: 87.6k, 2048: 105k, 4096: 112k, 8192: –; out of memory at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow
shadow_walled_5_sdf	Lodos, cone	64: 13.6k, 128: 22.2k, 256: 33.9k, 512: 55.7k, 1024: 73.3k, 2048: 91.9k, 4096: 106k, 8192: 113k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped
shadow_walled_5_sdf	Lodos, ray	64: 14.5k, 128: 23.8k, 256: 36.1k, 512: 59.6k, 1024: 75.8k, 2048: 91.7k, 4096: 104k, 8192: 110k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped
shadow_walled_5_sdf	MuJoCo Warp	64: 6.3k, 128: 9.59k, 256: 13.1k, 512: 16k, 1024: 17k, 2048: 17.4k, 4096: 17.4k, 8192: –; out of memory at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow
nist_threading_m8	Lodos, contact	64: 116k, 128: 216k, 256: 396k, 512: 666k, 1024: 980k, 2048: 1.36M, 4096: 1.67M, 8192: 1.74M, 16384: 1.73M; dropped at 16384; 0 non-finite, 0 escaped
nist_threading_m8	Lodos, cone	64: 108k, 128: 202k, 256: 365k, 512: 658k, 1024: 998k, 2048: 1.43M, 4096: 1.83M, 8192: 1.94M, 16384: 1.92M; dropped at 16384; 0 non-finite, 0 escaped
nist_threading_m8	Lodos, ray	64: 95k, 128: 180k, 256: 329k, 512: 579k, 1024: 958k, 2048: 1.44M, 4096: 1.91M, 8192: 1.97M, 16384: 1.94M; dropped at 16384; 0 non-finite, 35 escaped
nist_threading_m8	MuJoCo Warp	64: 35.5k, 128: 42.8k, 256: 47.7k, 512: 50.9k, 1024: 52k, 2048: 51.8k; dropped at 2048; 0 non-finite, 0 escaped
cube_walled_20_polytope	Lodos, contact	64: 41.7k, 128: 76.3k, 256: 134k, 512: 217k, 1024: 272k, 2048: 328k, 4096: 365k, 8192: 384k, 16384: 394k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
cube_walled_20_polytope	MuJoCo Warp	64: 40.9k, 128: 72k, 256: 120k, 512: 179k, 1024: 212k, 2048: 236k, 4096: 252k, 8192: 252k, 16384: 256k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
cube_walled_20_sdf	Lodos, contact	64: 23.5k, 128: 38k, 256: 60.6k, 512: 80.9k, 1024: 98.3k, 2048: 111k, 4096: 119k, 8192: –; out of memory at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow
cube_walled_20_sdf	Lodos, cone	64: 33.4k, 128: 58.7k, 256: 94.5k, 512: 152k, 1024: 188k, 2048: 224k, 4096: 247k, 8192: 257k, 16384: 263k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
cube_walled_20_sdf	Lodos, ray	64: 50k, 128: 86.3k, 256: 133k, 512: 211k, 1024: 264k, 2048: 309k, 4096: 337k, 8192: 349k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped
cube_walled_20_sdf	MuJoCo Warp	64: 27.1k, 128: 44.1k, 256: 67.8k, 512: 90k, 1024: 98.7k, 2048: 105k, 4096: 107k, 8192: 106k; dropped at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow
arm_walled_20_polytope	Lodos, contact	64: 34.9k, 128: 58.6k, 256: 91.3k, 512: 128k, 1024: 146k, 2048: 162k, 4096: 172k, 8192: 176k, 16384: 178k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
arm_walled_20_polytope	MuJoCo Warp	64: 21.6k, 128: 30.7k, 256: 38.9k, 512: 44.2k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped; 1 redone after an overflow
arm_walled_20_polytope	MuJoCo Warp, SAP	64: 21.6k, 128: 30.3k, 256: 38.5k, 512: 43.8k, 1024: 45.5k, 2048: 46.3k, 4096: 46.3k; dropped at 4096; 0 non-finite, 0 escaped; 2 redone after an overflow
arm_walled_20_sdf	Lodos, contact	64: 41.3k, 128: 70.8k, 256: 110k, 512: 167k, 1024: 202k, 2048: 232k, 4096: 249k, 8192: 257k, 16384: 261k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped

scene	configuration	world-steps/s by batch size; how the ladder ended; worlds lost at the best
`arm_walled_20_sdf`	Lodos, cone	64: 38.7k, 128: 65.8k, 256: 102k, 512: 158k, 1024: 194k, 2048: 227k, 4096: 247k, 8192: 255k, 16384: 259k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
`arm_walled_20_sdf`	Lodos, ray	64: 57.3k, 128: 96.2k, 256: 140k, 512: 212k, 1024: 269k, 2048: 317k, 4096: 349k, 8192: 362k, 16384: 366k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
`arm_walled_20_sdf`	MuJoCo Warp	64: 32k, 128: 47.7k, 256: 64.7k, 512: 78k, 1024: 81.8k, 2048: 83.5k, 4096: 84k, 8192: 83.7k; dropped at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow
`shadow_walled_20_polytope`	Lodos, contact	64: 1.08k, 128: 1.83k, 256: 2.68k, 512: 3.31k, 1024: 3.24k; dropped at 1024; 0 non-finite, 0 escaped
`shadow_walled_20_polytope`	MuJoCo Warp	64: 960, 128: 1.56k, 256: –; out of memory at 256; 0 non-finite, 0 escaped
`shadow_walled_20_polytope`	MuJoCo Warp, SAP	64: 958, 128: 1.51k, 256: 2.14k, 512: 2.59k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped
`shadow_walled_20_sdf`	Lodos, contact	64: 860, 128: 1.38k, 256: 2.11k, 512: 2.88k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped
`shadow_walled_20_sdf`	Lodos, cone	64: 631, 128: 1.05k, 256: 1.64k, 512: 2.3k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped
`shadow_walled_20_sdf`	Lodos, ray	64: 693, 128: 1.14k, 256: 1.73k, 512: 2.36k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped
`shadow_walled_20_sdf`	MuJoCo Warp	64: 626, 128: –; failed at 128; 0 non-finite, 0 escaped

Table 7: Every ladder: each batch size's world-steps/s, from 64 doubled, how it ended, and the worlds the best run lost.

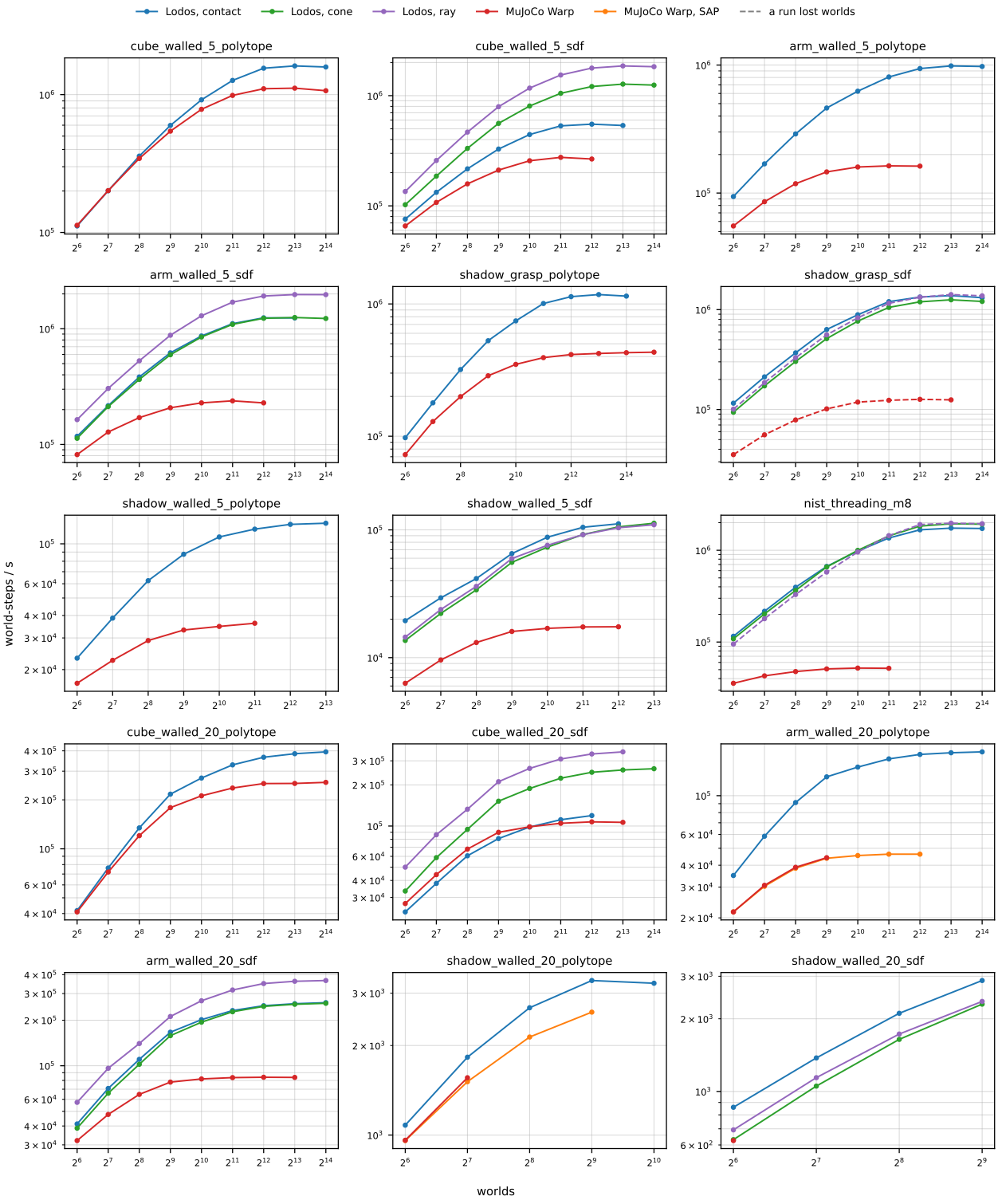


Figure 1: World-steps per second against the batch size, each configuration's ladder; dashed where a run lost worlds.